1839

Vulcanized rubber

American inventor Charles Goodyear discovered he could make rubber stronger by heating it with sulfur. The process—called vulcanization, after Vulcan, the Roman god of fire—makes rubber less sticky, increasing its practical use. Car tires are made of vulcanized rubber.

Goodyear experiments with rubber

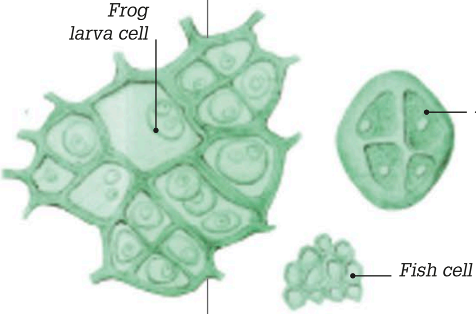
1815–1852 ADA LOVELACE

Ada Lovelace, the daughter of English poet Lord Byron, was a mathematician who worked with English inventor Charles Babbage on his Analytical Engine, a general-purpose computing machine (see p.124). Lovelace wrote the world's first "computer program" in 1843 and the modern programming language Ada is named after her.

Lovelace predicted that a computer could do more than numerical calculations—perhaps even compose music.

1840

1845



Theodor Schwann's drawings of cells

1839

Cell theory

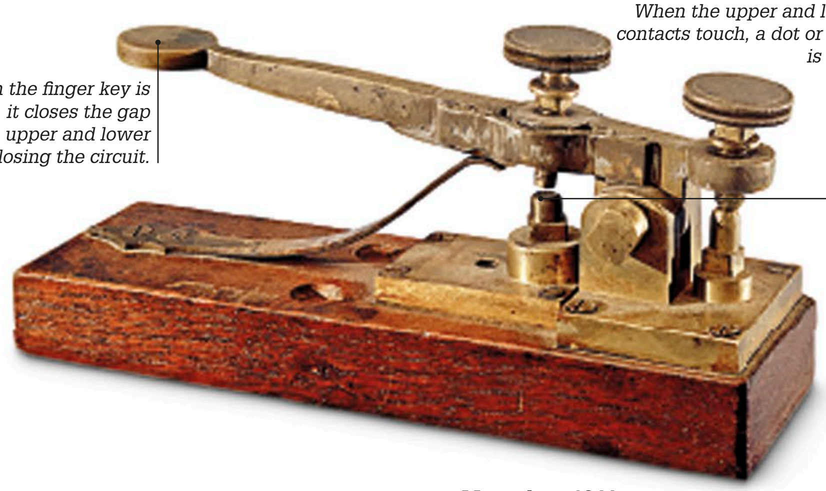
Theodor Schwann, a German physiologist, published a book called *Microscopical Researches*, in which he showed that all living things—animals and plants—are made from cells, and that the cell is the basic unit of life. He had worked out his ideas in collaboration with German botanist Matthias Schleiden.

Communication
See pages 150–151

1844

Down the wire

American inventor Samuel Morse sent a long-distance telegraph message down an electric wire from Washington, D.C. to Baltimore. He tapped out a message using Morse code, a system of dots and dashes he had devised to spell out the letters of the alphabet (see p.151).



Morse key, 1844

"What hath God wrought?"

First telegraph message, May 24, 1844